Review Questions on Chapter 2: Relational Database

1. Explain the difference between a candidate key, primary key, and foreign key with suitable examples.
2. Define the relational algebra operations "selection" and "projection" and give an example for each.
3. What is the difference between a relation *schema* and a relation *instance*?
4. Describe the differences in meaning between the terms *relation* and *relations chema*.
5. List two reasons why null values might be introduced into the database.

branch(*branch name, branch city, assets*)
customer (*customer name*, *customer street*, *customer city*)
loan (*loan number, branch name, amount*)
borrower (*customer name*, *loan number*)
account (*account number*, *branch name, balance*)
depositor (*customer name*, *account number*)

Figure 2.15 Banking database

6. Consider the bank database of Figure 2.15.
   i). What are the appropriate primary keys?
   ii). Given your choice of primary keys, identify appropriate foreign keys.
7. Using relational algebra, write a query to find the names of students enrolled in the 'Computer Science' department.
8. Given the relations:
   Employee(emp_id, emp_name, dept_id) and
   Department(dept_id, dept_name)
   Now, write the relational algebra expression
   i). To get employee names with their department names.
   ii). Write a relational algebra query to list students older than 20 years.
9. Apply projection and selection together to display only the names of employees who work in the 'Finance' department.

10. Compare relational algebra and SQL in terms of their query approach and usage in databases.
11. Analyze the difference between Cartesian product and natural join in relational algebra. Provide an example where natural join is more useful.
12. Examine how division (÷) operator in relational algebra can be used to answer "for all" type queries.

Note: Follow the lecture topics and last of the book chapter questions.